

Thermodynamic Phase Stability and Local Database Workflows for Cu/Ga Bulk and Surface Structures

A Computational Chemistry Master Thesis Based on OnePiece, ASE and
pandas DataFrames

Codex

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Abstract

This thesis investigates a local computational chemistry dataset of Cu/Ga bulk, oxide-derived bulk, cluster and surface structures stored as pandas HDF files. The dataset combines density-functional-theory-derived energies, relaxed atomic structures, geometric descriptors, Bader-like charge descriptors, surface coverages, reference energies and thermodynamic correction terms. The work has two connected aims. First, it develops a reproducible analysis of bulk and surface phase stability as a function of temperature and the gas ratio $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$. Second, it designs a local user-interface workflow for browsing, filtering, reviewing and reusing such atomistic data in a way that resembles high-quality materials databases while remaining entirely local and file-based.

The computational workflow is implemented in Python with pandas, NumPy, SymPy, ASE and Plotly/Matplotlib. The input consists of eight HDF files containing 274 calculation records and 90 original columns. Additional context columns are derived for record type, reference role, phase labels, quality flags, atom counts and phase-diagram usage. Bulk stability is evaluated by converting symbolic free-energy expressions to numerical functions over a grid of temperature and $\log_{10}(p_{\text{H}_2\text{O}}/p_{\text{H}_2})$. Surface stability is evaluated by correcting surface free energies with chemical potential terms and comparing candidates per surface area. The resulting phase fields identify stable Cu-rich, intermediate Cu/Ga and Ga-rich regions in the chosen thermodynamic window. Individual Miller-index analyses show distinct surface-stability regions for (100), (110), (111) and (211) surfaces.

The thesis also presents PFUI, a local Python package prototype for a materials database front end. PFUI treats OnePiece-like data objects, HDF files and pandas DataFrames as interchangeable sources, displays ASE-aware records, supports image-path columns, and introduces a Controlroom tab for filtering, row inclusion states, reference discovery and command-like notebook operations. This combination of computational analysis and interface design provides a reproducible foundation for future adsorption-energy workflows in which clean surface references and adsorbate references are resolved directly from the local dataset.

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Chapter 1

Introduction

The analysis of alloy surfaces is central to modern computational chemistry, heterogeneous catalysis and materials design. Copper–gallium systems are particularly interesting because surface enrichment, oxide-derived thermodynamics and adsorbate binding may all change with temperature, gas composition and surface orientation. A single relaxed structure is therefore rarely sufficient. Instead, meaningful interpretation requires a database of bulk, surface, reference and candidate calculations that can be filtered, compared and transformed into thermodynamic observables.

Large public platforms such as the Materials Project and OQMD demonstrate how valuable searchable materials databases can be. In a research project, however, data often exists first as local calculation folders, HDF files and notebooks. The scientific problem is then not only to compute a phase diagram, but to make the data workflow auditable: which rows were used, which calculations were excluded, which reference surface was selected, which energy expression was evaluated, and which notebook or script produced a figure.

This thesis addresses that local workflow. The dataset studied here was exported from a OnePiece-like computational chemistry package into pandas HDF files. The files contain energies, ASE atomic structures, formulas, surface descriptors, chemical potentials and derived thermodynamic quantities. The central scientific task is to construct Cu/Ga phase diagrams as functions of temperature and the water/hydrogen gas ratio. The central software task is to design an interface that allows the same calculations to be inspected, filtered and repeated without forcing the user to edit notebook code for every question.

1.1 Research Questions

The thesis is guided by four questions:

Q1. How can OnePiece HDF files be represented as a chemically meaningful pandas

DataFrame while preserving provenance and structure context?

- Q2.** Which bulk and surface phases are stable in a thermodynamic window defined by T and $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$?
- Q3.** Which column profiles and derived descriptors are needed for different local database pages such as overview, surface stability, phase diagrams, reference matching and quality control?
- Q4.** How can notebook commands be translated into an efficient local user interface for filtering, excluding and exporting database rows?

1.2 Contributions

The main contributions are:

- a reproducible pandas/ASE workflow for reading and combining Cu/Ga HDF files;
- tutorial notebooks explaining the workflow for chemistry students with Excel-level table experience;
- bulk and surface phase diagrams across temperature and gas ratio;
- a local Streamlit-based PFUI prototype with a Controlroom tab;
- a column-profile system that selects relevant columns for different scientific pages;
- a foundation for future adsorption-energy calculations based on automatic reference matching through row names and structure metadata.

Chapter 2

Theoretical Background

2.1 Density Functional Theory and Atomistic Databases

The electronic energies in the dataset originate from density functional theory (DFT), in which the electronic many-body problem is mapped to an effective single-particle problem through the electron density. The Kohn–Sham formalism provides the practical foundation for most modern DFT calculations [1]. In typical surface and alloy studies, the exchange correlation functional is approximated, for example with a generalized gradient approximation such as PBE [2].

For this thesis, the exact electronic-structure input settings are treated as part of the provenance stored in the local database. The important point is that the DFT calculations produce total energies, relaxed atomic structures and diagnostic quantities such as residual forces. These values are then reorganized into a DataFrame for chemical analysis.

2.2 Formation Energies

For a bulk alloy phase containing n_{Cu} copper atoms and n_{Ga} gallium atoms, a formation energy can be written in a generic form as

$$\Delta E_{\text{form}} = E_{\text{Cu}_x\text{Ga}_y} - n_{\text{Cu}}\mu_{\text{Cu}} - n_{\text{Ga}}\mu_{\text{Ga}}. \quad (2.1)$$

The dataset contains several formation-energy variants, including per-atom, per-alloy and per-volume forms. The phase-diagram workflow uses `formation_energy_per_atom` for the bulk analysis. Some entries are plain numerical values, while others are symbolic expressions containing temperature, gas pressure terms and molecular reference quantities.

2.3 Oxygen Chemical Potential

The gas-dependent thermodynamic correction is expressed through an effective oxygen chemical potential based on the water/hydrogen ratio:

$$\mu_O = E_{\text{H}_2\text{O}} - E_{\text{H}_2} - T(S_{\text{H}_2\text{O}} - S_{\text{H}_2}) + k_B T \ln \left(\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} \right). \quad (2.2)$$

The notebooks use

$$E_{\text{H}_2} = -6.737 \text{ eV}, \quad (2.3)$$

$$E_{\text{H}_2\text{O}} = -12.062 \text{ eV}, \quad (2.4)$$

$$S_{\text{H}_2} = 1.5044 \times 10^{-3} \text{ eV K}^{-1}, \quad (2.5)$$

$$S_{\text{H}_2\text{O}} = 2.2198 \times 10^{-3} \text{ eV K}^{-1}, \quad (2.6)$$

$$k_B = 8.617333262145 \times 10^{-5} \text{ eV K}^{-1}. \quad (2.7)$$

The plotted vertical axis is $\log_{10}(p_{\text{H}_2\text{O}}/p_{\text{H}_2})$, whereas Equation 2.2 uses the natural logarithm. Therefore, the numerical workflow converts

$$\ln \left(\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} \right) = \ln(10) \log_{10} \left(\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} \right). \quad (2.8)$$

2.4 Surface Free Energies

For surfaces, energies must be normalized by area and corrected by reference chemical potentials. The working expression used for the surface diagrams is

$$G_{\text{surf}}^{\text{corr}} = \frac{G_{\text{stored}} + \Delta n_{\text{Ga}}(\mu_{\text{Ga}}^{\text{ref}} - \mu_{\text{Ga}}) + \Delta n_{\text{Cu/Zn}}(\mu_{\text{Cu/Zn}}^{\text{ref}} - \mu_{\text{Cu/Zn}})}{A}. \quad (2.9)$$

Here A is the surface area. The exact column names in the dataset include `form_G`, `Area`, `delta_Ga`, `delta_Cu`, `delta_Zn`, `mu_Ga`, `mu_Cu` and `mu_Zn`.

Chapter 3

Data and Software Infrastructure

3.1 Input HDF Files

The local Cu/Ga dataset consists of eight HDF files loaded with `pd.read_hdf(filename, key="df")`. They include bulk, oxide-derived bulk, cluster and individual surface datasets. After combination, the working cache contains 274 rows and 90 original columns. The row distribution is shown in Figure 3.1.

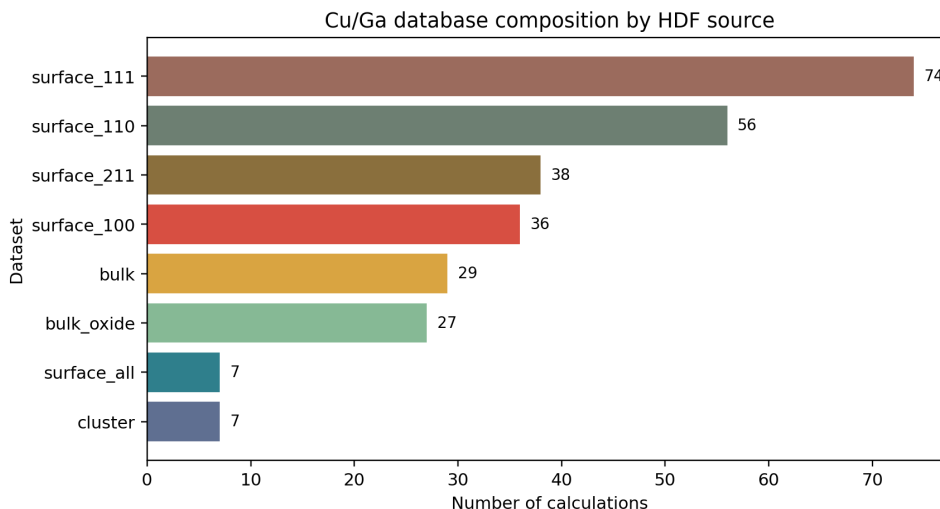


Figure 3.1: Number of calculation records per HDF source in the combined Cu/Ga dataset.

3.2 Python Data Model

The workflow uses pandas DataFrames as the central data representation [3]. Each row corresponds to one calculation record, and columns store energies, formulas, atom counts, structure objects, reference paths and descriptors. ASE provides the atomistic

Table 3.1: Summary of the combined local Cu/Ga database.

Dataset	Rows	Unique formulas	Mean $f_{\max}$
bulk	29	24	0.0066
bulk_oxide	27	22	0.0045
cluster	7	1	0.0079
surface_100	36	30	0.0071
surface_110	56	37	0.0070
surface_111	74	33	0.0064
surface_211	38	30	0.0073
surface_all	7	1	0.0079

structure representation [4]. NumPy is used for numerical arrays and grid evaluation [5], and Matplotlib/Plotly are used for figures and interactive surfaces [6].

Several derived columns are added for UI and analysis:

- `row_uid`: stable identifier based on HDF source and source row;
- `record_type`: bulk, surface, cluster or generic record;
- `is_clean`: whether the row appears to be a clean reference;
- `is_adsorbate_like`: whether the name suggests an adsorbate;
- `reference_role`: candidate, clean surface reference, bulk reference or adsorbate candidate;
- `n_atoms`, `energy_per_atom` and `quality_flag`.

3.3 Tutorial Notebooks

The thesis figures are derived from the tutorial notebooks and scripts created for this project:

- `cuga_mu0_temperature_phase_diagram.ipynb`;
- `00_hdf_onepiece_pandas_ase_intro.ipynb`;
- `01_bulk_phase_table_from_hdf.ipynb`;
- `02_surface_phase_tables_from_hdf.ipynb`;
- `03_multiplot_transition_tables.ipynb`.

The notebooks are intentionally written as teaching material for chemists who are familiar with tables from Excel but are still learning pandas.

Chapter 4

Methods

4.1 Data Loading and Cleaning

The HDF files are read into pandas and concatenated into a single local cache. Three provenance columns are inserted: `dataset`, `source_hdf` and `source_row`. These columns ensure that every row in a derived plot can be traced back to the original HDF source.

Some columns contain mixed types. For example, `formation_energy_per_atom` can contain a number in one row and a symbolic expression in another. This is chemically meaningful but challenging for generic table renderers. Therefore, numerical versions such as `formation_energy_per_atom_numeric` are derived where possible while the symbolic text is preserved for auditability.

4.2 Quality Screening

The maximum residual force `fmax` is used as a simple relaxation-quality indicator. The distribution is shown in Figure 4.1. The practical UI threshold used in the Controlroom is $f_{\max} < 0.05$ eV/Å for quick screening. This threshold is not a universal physical law; it is a local review rule that can be adjusted by the user.

4.3 Bulk Phase-Diagram Algorithm

The bulk phase diagram is computed as follows:

1. define a temperature grid from 300 to 1100 K;
2. define a gas-ratio grid from $\log_{10}(p_{\text{H}_2\text{O}}/p_{\text{H}_2}) = -16$ to 4;
3. convert each bulk free-energy expression to a numerical function;

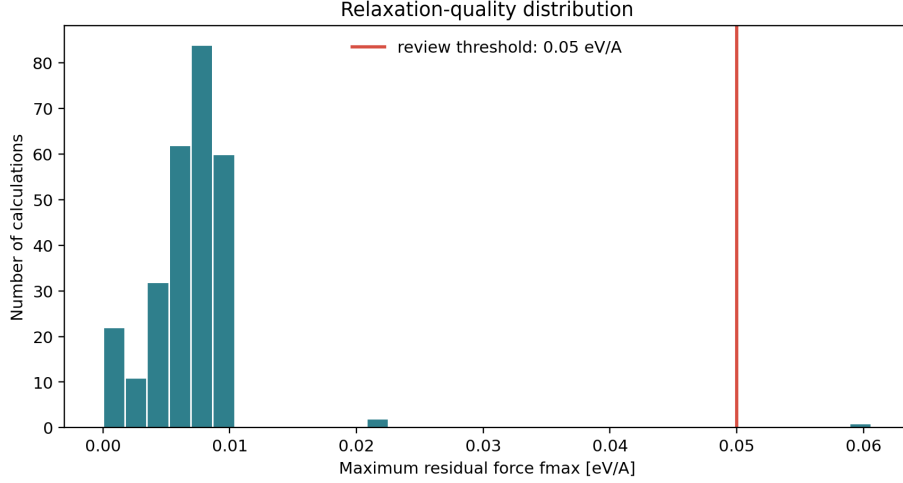


Figure 4.1: Distribution of maximum residual forces in the combined Cu/Ga dataset. The red line marks the default review threshold used in the PFUI Controlroom.

4. evaluate all phase energies on the grid;
5. select the phase with minimum energy at every grid point.

This is equivalent to a large spreadsheet in which each cell contains the energy of a candidate phase under one condition, and the stable phase is found by taking the row-wise minimum across candidates.

4.4 Surface Phase-Diagram Algorithm

For each Miller index, the surface table is evaluated separately. Surface candidates are corrected using Equation 2.9, normalized by area, and compared on the same temperature/gas-ratio grid. Individual (100), (110), (111) and (211) maps are then combined into a multi-panel view.

Chapter 5

Results

5.1 Bulk Energetics

Figure 5.1 summarizes the bulk formation energies as a function of Ga content. The dataset contains Cu-rich phases, intermediate Cu/Ga phases and Ga-rich oxide-derived candidates. The wide spread in formation energy reflects the different chemical references and thermodynamic terms represented in the combined HDF files.

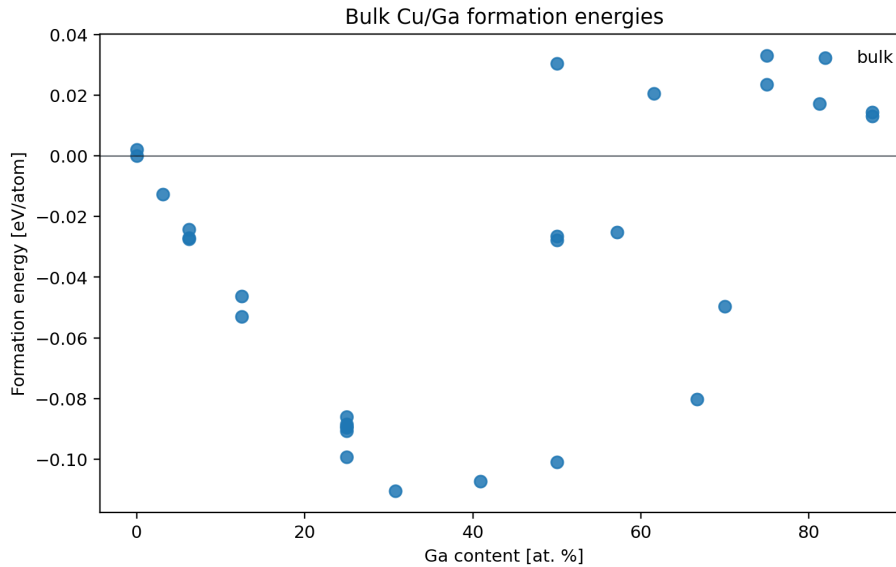


Figure 5.1: Bulk formation energies as a function of gallium content for the bulk and oxide-derived bulk datasets.

5.2 Bulk Phase Diagram

Figure 5.2 shows the bulk phase map generated by the teaching notebook. At each point of the temperature/gas-ratio grid, the color indicates the phase with the lowest

free energy. The phase fields are therefore not fitted boundaries; they are the result of direct energy comparison across all candidate expressions.

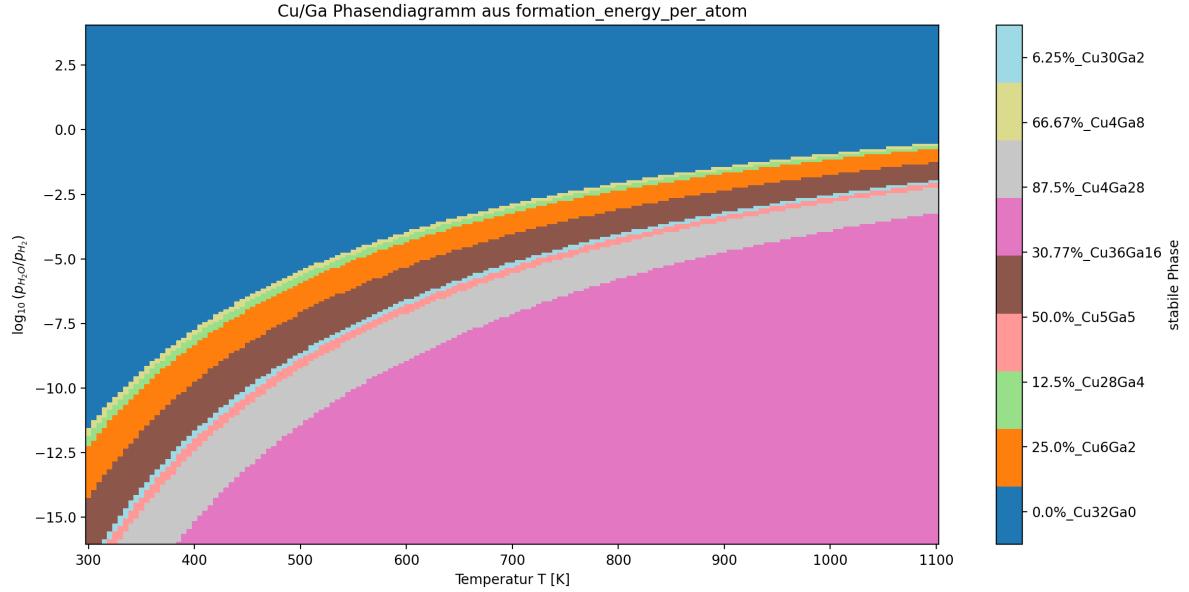


Figure 5.2: Bulk Cu/Ga phase diagram as a function of temperature and $\log_{10}(p_{\text{H}_2\text{O}}/p_{\text{H}_2})$. The figure is generated from `cuga_mu0_temperature_phase_diagram.ipynb`.

The largest stable bulk regions in the computed window are summarized in Table 5.1. The Cu-rich reference and a Ga-rich phase dominate large fractions of the evaluated grid. Intermediate phases occupy smaller transition regions.

Table 5.1: Representative stable bulk phases in the computed window.

Name	Formula	Ga [%]	Stable grid [%]	$G_{\min}$
Cu-bulk-Cu32	Cu32	0.0	38.91	0.000
Cu-bulk-Cu4Ga28_a	Cu4Ga28	87.5	38.30	-4.039
Cu-bulk-CuGa2	Cu4Ga8	66.7	7.65	-0.379
Cu-bulk-Cu9Ga4	Cu36Ga16	30.8	6.16	-0.120
Cu-bulk-Cu6Ga2	Cu6Ga2	25.0	4.70	-0.050

5.3 Surface Stability

Figure 5.3 shows corrected surface free energies as a function of Ga surface coverage. The comparison highlights why surface phase diagrams must be evaluated per Miller index: the same nominal coverage does not imply the same stability across different facets, because area, slab geometry and reference terms differ.

The individual surface phase diagrams are shown in Figure 5.4. Each panel was computed by comparing the corrected surface free energies of candidates belonging to

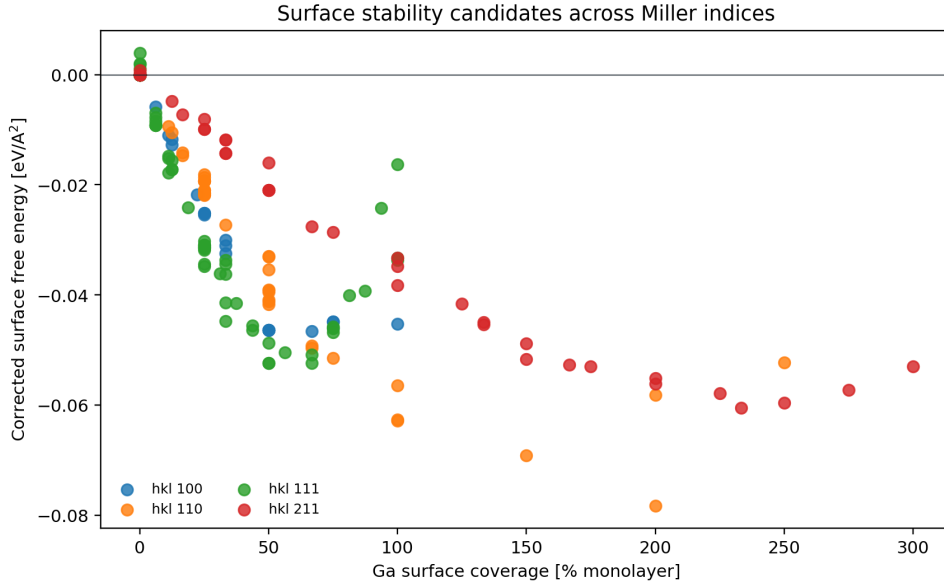


Figure 5.3: Corrected surface free energies per area as a function of Ga monolayer coverage for the surface datasets.

the same Miller index.

5.4 Combined Transition Fields

Figure 5.5 combines the bulk-derived and surface-derived transition fields in one view. This figure is useful for comparing which candidate phases dominate the thermodynamic window and where transitions occur. The accompanying extended CSV table stores rank, panel, phase identity, stability fraction, temperature span, gas-ratio span and energy source.

Figure 5.6 ranks the largest stability fields by grid fraction. This ranking is not a thermodynamic phase fraction; it only measures how much of the selected computational window is occupied by a phase.

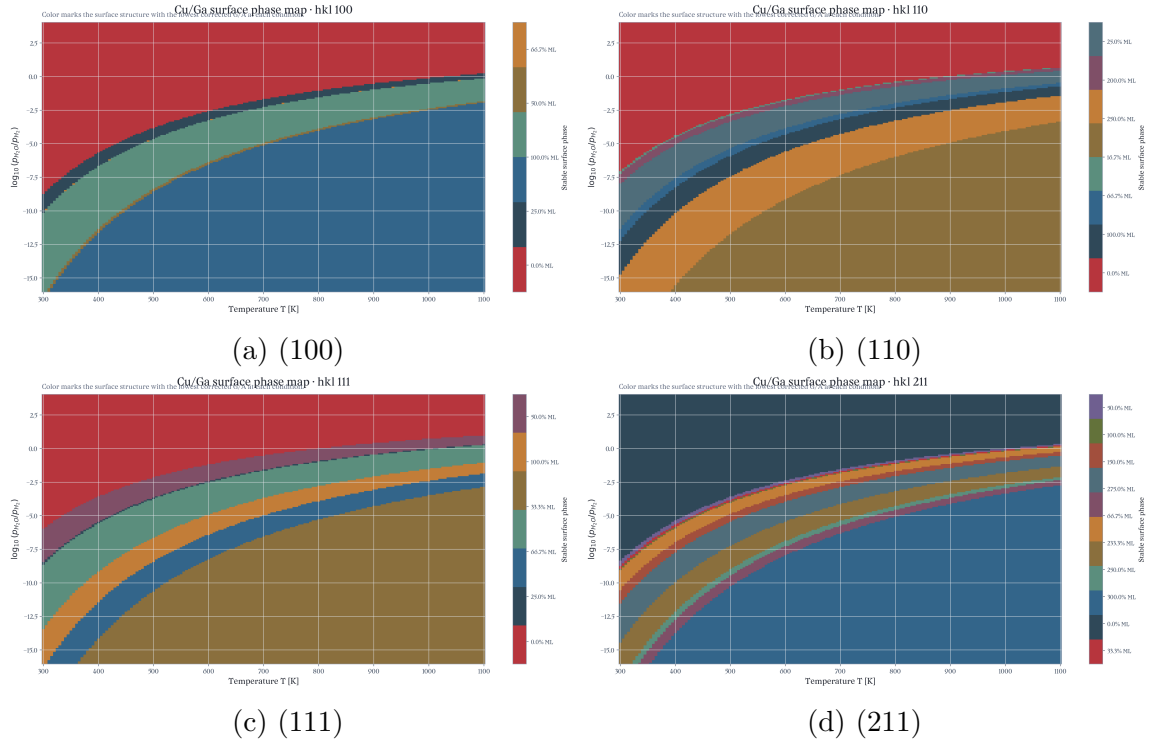


Figure 5.4: Individual Cu/Ga surface phase diagrams for four Miller indices. Colors indicate the stable surface candidate at each thermodynamic condition.

Cu/Ga transition-phase multiplot: bulk and individual surfaces

Each panel shows the thermodynamic phase with the lowest corrected free energy at each T and pH2O/pH2 condition.

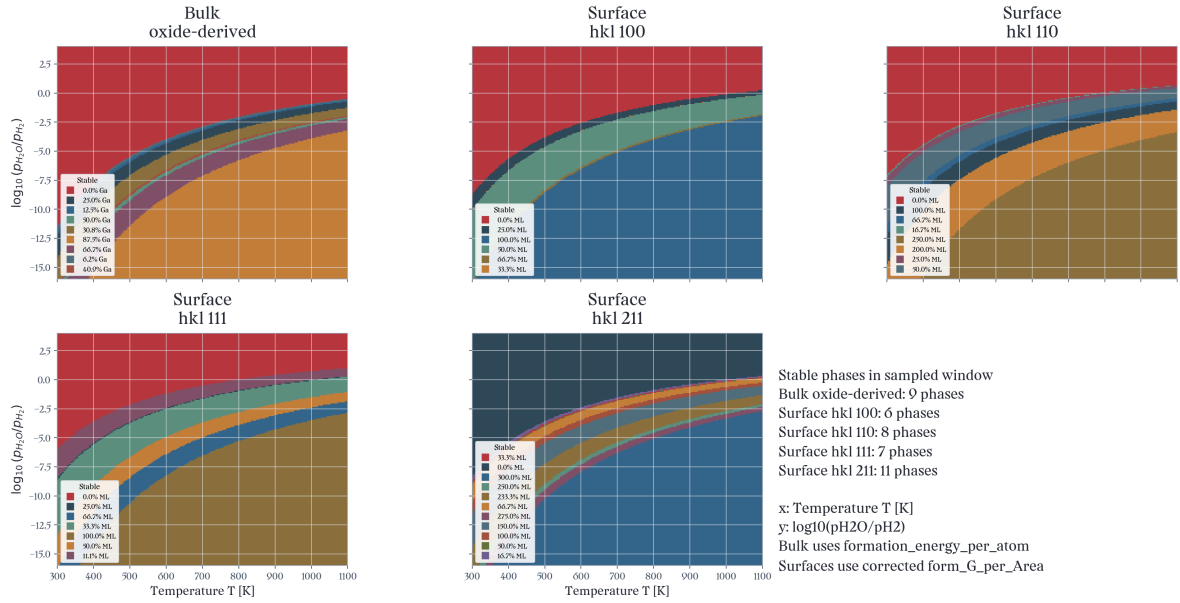


Figure 5.5: Combined transition-phase multiplot for bulk and surface panels.

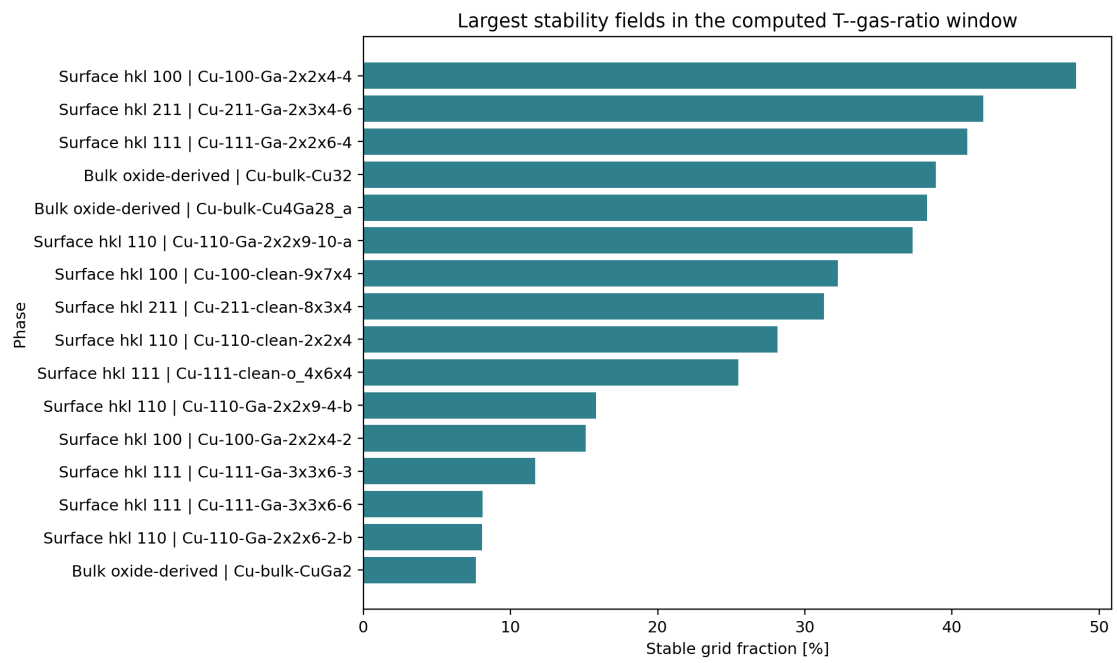


Figure 5.6: Largest stability fields in the computed temperature and gas-ratio window.

Chapter 6

Local User Interface Design

6.1 Motivation

Computational chemistry datasets are rarely static. New calculations are added, some rows become invalid, reference definitions evolve and derived energies must be recalculated. A notebook alone is excellent for reproducibility, but less efficient for repeated row review. PFUI was therefore developed as a local database front end that keeps the DataFrame model visible while making common operations clickable.

6.2 PFUI Architecture

PFUI accepts three source types:

- direct pandas DataFrames;
- HDF files loaded with `pd.read_hdf`;
- OnePiece-like objects exposing `to_dataframe()`, `.dataframe` or `.df`.

The UI contains Records, Controlroom, Visualize and Schema tabs. The Records tab shows row details and column profiles. The Visualize tab provides quick scatter and histogram views. The Schema tab documents column types and samples.

6.3 Controlroom Tab

The Controlroom translates notebook operations into UI commands. It supports:

- text include/exclude filters;
- row states: included, excluded, review and reference;

- materials facets such as dataset, HDF source, Miller index and slab size;
- numerical filters for energies, forces and surface descriptors;
- commands for clean references, adsorption-like rows, quality problems, phase-diagram candidate tables and column reviews.

This design follows the spirit of a calculation control room: rows are not hidden by accident, and every active subset can be exported as a CSV file.

6.4 Column Profiles

The same DataFrame should not be displayed with the same columns on every page. PFUI therefore defines column profiles for different scientific tasks:

Profile	Purpose
Overview	identity, formula, energy and quality scan
Surface Stability	surface geometry, coverage, area and corrected energies
Phase Diagram	thermodynamic inputs and phase labels
References	clean surfaces, bulk references and adsorption candidates
Quality	convergence, force and missing-data review
Structure	cell parameters, atom count and descriptor context
Provenance	source HDF, source row, local path and calculation metadata

Chapter 7

Future Adsorption-Energy Workflow

The next planned workflow is adsorption-energy analysis. For a target adsorbate–surface row, an adsorption energy can be expressed as

$$E_{\text{ads}} = E_{\text{adsorbate+surface}} - E_{\text{clean surface}} - n_{\text{ads}} E_{\text{adsorbate reference}}. \quad (7.1)$$

The key database challenge is reference resolution. The clean reference surface should be found from the same dataset, preferably by matching the **Name** column, Miller index, slab size and composition. PFUI therefore already derives `reference_role`, `reference_key`, `is_clean`, `is_adsorbate_like` and `adsorbate_guess`.

Ambiguous references should not be accepted silently. If multiple clean surfaces match one adsorbate row, the UI should mark the row as **review**. This is important because a wrong reference can shift adsorption energies by much more than typical trends between surface sites.

Chapter 8

Conclusions

This thesis demonstrates a local computational chemistry workflow for Cu/Ga bulk and surface datasets stored in OnePiece-style pandas HDF files. The data were combined into a 274-row working database, enriched with context columns and used to compute thermodynamic phase diagrams across temperature and $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$. Bulk and surface candidates show distinct stability fields, and the surface analysis confirms that each Miller index must be treated individually.

Beyond the numerical phase diagrams, the work emphasizes the importance of local database usability. The PFUI prototype provides a bridge between notebooks and a research database interface. It allows calculation rows to be filtered, reviewed, marked as references or excluded, and exported for further analysis. The column-profile design reduces cognitive load by showing the columns relevant to the current scientific task rather than presenting all available metadata at once.

The resulting workflow is reproducible, local and extensible. It is ready for future adsorption-energy calculations in which clean surfaces and molecular references are resolved directly from the dataset and every derived energy can be traced back to the source HDF rows.

Appendix A

Reproducibility Notes

The main thesis figures were generated with:

```
python thesis/build_thesis_figures.py
```

The source notebooks are located in:

```
notebooks/  
notebooks/onepiece_phase_tutorial/
```

The combined PFUI dataset cache is:

```
notebooks/phase_diagram_outputs/cuga_full_dataset.pkl
```

The PFUI app for browsing the complete local dataset is launched with:

```
python3 -m streamlit run examples/cuga_full_streamlit.py \  
--server.headless true --server.port 8502
```


Appendix B

Important Generated Files

- `thesis/main.tex`: this thesis.
- `thesis/build_thesis_figures.py`: figure-generation script.
- `thesis/figures/`: figures used by the thesis.
- `thesis/tables/`: CSV tables used for summaries.
- `src/pfui/materials_columns.py`: column-profile and derived descriptor logic.

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